Finite-Length Switching and Limitations of Recursive Constructions in Binary Deletion Channels

No.50 | Exact finite certificates, a failed proof strategy, and a bounded claim

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Preprint v1.0 | November 2, 2026 | All rights reserved. This research note combines exact finite-length computer-assisted results with a documented failed generalization. Independent AI audit: **PASS for the finite-length results** (audit outcome supplied by the author). Windows PowerShell reproduction of v0.2.1: **PASS** (author-reported execution on November 2, 2026). Novelty has not been conclusively established. This preprint has not undergone human peer review. The conjectured unbounded growth in switching complexity remains unproved. Failure of the proposed insertion strategy does not refute that conjecture.

Abstract

For binary inputs of equal length transmitted through an independent deletion channel, we consider the minimum total-variation distance between the resulting one-trace output laws, as a function of the deletion probability. We establish the exact lower envelopes for input lengths four through eight. The interior numbers of switches between distinct optimal distance polynomials are, respectively, $1, 1, 2, 3, 3$ for lengths $4, 5, 6, 7, 8$. The certificates use integer subsequence counts, coefficientwise domination, exact rational Bernstein inequalities, and isolation of the unique roots of adjacent candidate differences. A separate exhaustive example at deletion probability $4/5$ shows that inserting one common bit at arbitrary independently chosen positions into either of the two optimal length-seven pairs never gives an optimal length-eight pair. However, another length-seven optimizer can be inserted to obtain a length-eight optimizer at a *different* deletion probability: the counterexample negates same-parameter preservation, not the broader possibility of migrating optimal intervals. The attempted argument for an unbounded number of switches therefore remains unsuccessful. Previous subsequence-deck techniques and the elementary certification method are not claimed as new.

1. Scope, claim categories, and prior work

The motivation comes from OpenAI Math family #122 and its work on worst-case trace reconstruction [1]. That work concerns, in part, asymptotic one-trace indistinguishability for fixed channel parameters. Here we study a finite-length *lower envelope over the deletion parameter*. The underlying subsequence counting identity and k-deck equivalences are classical tools; see, for example, [2,3]. We do **not** claim an exhaustive priority search, or that each finite value herein is new to the literature.

We distinguish throughout:

1. **Proved, computer-assisted finite results:** full-parameter classifications at $n = 4, 5, 6, 7, 8$ and a specific insertion counterexample.
2. **Elementary auxiliary statement:** fixed-length neighborhood stability, included without a novelty claim.
3. **Failed proof approach:** extending an optimizer by one common inserted bit is not guaranteed to preserve optimum at the *same* deletion parameter.
4. **Unproved conjecture:** the number of switches becomes arbitrarily large as word length grows.

2. Deletion channel and distance polynomials

Write $\mathcal{D}_q(x)$ for the probability law of the output from input $x \in \{0,1\}^n$ when each symbol is independently deleted with probability q. Let $N_x(w)$ count the deletion masks producing subsequence w. When $|w| = k$,

$$\Pr(\mathcal{D}_q(x) = w) = N_x(w)(1-q)^k q^{n-k}.$$

For distinct equal-length words x, y, define

$$\begin{aligned} D_{x,y}(q) &= \mathrm{TV}(\mathcal{D}_q(x), \mathcal{D}_q(y)) \\ &= \frac{1}{2}\sum_{k=0}^{n}(1-q)^k q^{n-k} \sum_{|w|=k} |N_x(w) - N_y(w)|. \end{aligned} \tag{1}$$

Set $a_k(x,y) = \frac{1}{2}\sum_{|w|=k} |N_x(w) - N_y(w)|$. Each a_k is an integer because the two collections of length-k subsequence counts have equal total mass $\binom{n}{k}$. For a vector $a = (a_0, \dots, a_n)$ put

$$P_{n,a}(q) = \sum_{k=0}^{n} a_k (1-q)^k q^{n-k}, \qquad \delta_n(q) = \min_{x \neq y} D_{x,y}(q). \tag{2}$$

Distinct word pairs can have exactly the same distance polynomial. Thus $S(n)$ denotes the number of *interior changes of the optimizing polynomial*, not the number of changes of the identity of an optimizing word pair.

3. Complete exact optimization for lengths four and five

Proposition 1 (computer-assisted). For every $q \in [0,1]$,

$$\boxed{\delta_4(q) = \delta_5(q) = \min\{A(q), B(q)\}}, \tag{3}$$

where $A(q) = (1-q)^2(q^2 - q + 1)$ and $B(q) = (1-q)^3(3q+1)$. There is exactly one interior switch, at $q = 3/4$. Hence $S(4) = S(5) = 1$.

The witnesses are `0010/0100` and `0110/1001` at $n = 4$; adding a common initial zero yields the displayed witnesses at $n = 5$. The enumerator checks all 120 and 496 distinct input pairs, on both intervals $[0, 3/4]$ and $[3/4, 1]$. Each of the **1,232 pair-by-interval inequalities** is proved by nonnegative exact rational Bernstein coefficients. The claim is computer-assisted, conditional on the finite enumerator, not a hand verification of 1,232 inequalities.

4. Exact lower envelopes for lengths six through eight

Theorem 2 (computer-assisted finite envelope). For $n = 6, 7, 8$, the exact function $\delta_n(q)$ over $0 \le q \le 1$ is the minimum of $P_{n,a}(q)$ over the coefficient vectors listed in Table 1. Their order on the open intervals of $(0,1)$ is the table order. The exact number of interior switches is

$$\boxed{S(6) = 2, \quad S(7) = 3, \quad S(8) = 3.} \tag{4}$$

Table 1. Optimal distance-polynomial vectors, with coefficients ordered as $(a_0, \dots, a_n)$.

n	candidate	representative unordered pair	coefficient vector a
6	0	`000100 / 001000`	$(0,0,1,2,4,2,1)$
6	1	`010110 / 011010`	$(0,0,1,2,3,3,1)$

n	candidate	representative unordered pair	coefficient vector a
6	2	`001101 / 010011`	$(0,0,0,4,6,6,1)$
7	0	`0000100 / 0001000`	$(0,0,1,3,6,6,3,1)$
7	1	`0010110 / 0011010`	$(0,0,1,3,5,6,4,1)$
7	2	`0011001 / 0100101`	$(0,0,0,4,8,12,7,1)$
7	3	`0110001 / 1000110`	$(0,0,0,0,21,19,7,1)$
8	0	`00001000 / 00010000`	$(0,0,1,3,9,9,9,3,1)$
8	1	`00101100 / 00110100`	$(0,0,1,3,7,10,9,5,1)$
8	2	`01011001 / 01100101`	$(0,0,0,4,7,16,18,8,1)$
8	3	`00110001 / 01000110`	$(0,0,0,0,21,38,26,8,1)$

The switch points are the unique roots in the specified rational isolating intervals of the following equations. All interval endpoints in Table 2 are multiples of 10^{-6}, except the exact root $1/2$.

Table 2. Algebraic switching points in increasing order.

n	index	defining equation $f(q)=0$	isolating interval for q
6	1	$2q-1$	$q=1/2$ exactly
6	2	$3q^3-5q^2+6q-3$	$(0.763072, 0.763073)$
7	1	$2q-1$	$q=1/2$ exactly
7	2	$2q^4-q^3-3q^2+6q-3$	$(0.742622, 0.742623)$
7	3	$10q^2+q-7$	$(0.788152, 0.788153)$
8	1	$5q^3-7q^2+6q-2$	$(0.544922, 0.544923)$
8	2	$2q^5+2q^4-6q^3+6q-3$	$(0.717875, 0.717876)$
8	3	$2q^3+3q^2+q-4$	$(0.826005, 0.826006)$

Exhaustive certificate argument. At $n=6,7,8$, the numbers of unordered pairs checked are respectively $2,016$, $8,128$, and $32,640$. Their numbers of distinct distance-coefficient vectors are respectively 260, $1,190$, and $5,835$. For lengths six and seven, *every* distance vector is coefficientwise greater than or equal to one of the candidate vectors in Table 1. At length eight, that statement holds for all but six distance vectors. Each exceptional vector is certified above one of the candidates over $[0,1/2]$ and above (possibly another) candidate over $[1/2,1]$, using nonnegative rational Bernstein coefficients. Hence no omitted polynomial can fall below the minimum of the Table 1 candidates.

For the table candidates themselves, the intervals outside the narrow rational isolating windows in Table 2 are certified by rational Bernstein comparisons to have the stated unique minimizing candidate. Each transition window has only its two adjacent candidates at or below the other candidates; for each adjacent difference, opposite endpoint signs and strictly positive derivative Bernstein coefficients certify **one and only one** crossing inside the window. At $q=1/2$, the adjacent difference vanishes exactly, while the adjacent open regions are certified separately. This proves that the candidate order, the precise algebraic roots, and the switch counts in (4) hold throughout the *whole parameter range*, not merely on a sampled grid.

All certification arithmetic is exact (`int` and `fractions.Fraction` in Python); `code/verify_envelope_n6_n8.py` generates the machine-checkable certificate `evidence/envelope_n6_n8.json`. This statement is a computer-assisted finite theorem, subject to independent software and priority audits.

Corollary (repaired v0.1 witness argument). At $n=7$ the four witness points $1/4, 3/5, 3/4, 4/5$ have **singleton and mutually disjoint sets of minimizing distance polynomials**, even though each point

may have multiple minimizing pairs. The verifier checks the complete winning sets; merely comparing one representative vector from each point would not suffice. The four witness polynomials already guarantee $S(7) \geq 3$, and Theorem 2 strengthens this to the exact equality $S(7) = 3$.

5. Specific failed recurrence and migrating optimizer

Proposition 3 (finite counterexample at a fixed parameter). At deletion probability $q = 4/5$, the only two minimizing unordered length-seven word pairs are

$$(0110001, 1000110), \qquad (0111001, 1001110).$$

Their optimum is $\delta_7(4/5) = 1677/78125$. For either pair, insert the same bit $b \in \{0, 1\}$ at independently selected positions of its two words, with eight positions per word. Across all $2 \times 2 \times 8 \times 8 = 256$ **operations** (not necessarily distinct results), the best distance is $8257/390625$, while the actual minimum at length eight is $\delta_8(4/5) = 7233/390625$, attained for example by `01011001/01100101`. Thus no such insertion preserves the *same-q* global optimum in this example. The candidate universal lemma of same-parameter one-bit insertion preservation is **false**.

Qualification from independent audit. The length-eight pair `01011001/01100101` is obtained by prepending `0` to the length-seven pair `1011001/1100101`, which is among the optimal *distance-polynomial class* at $q = 3/4$. At length eight the resulting pair is optimal at $q = 4/5$. Therefore one must not claim that inserted optimal pairs can **never** regain global optimality at other parameters. The audit suggested possible migration of optimality intervals; the given instance is illustrative, **not** a general recurrence theorem and not a proof that interval migration always holds.

The limitation is logical: failure of a particular *same-parameter extension* is not a counterexample to the conjecture that $S(n)$ is unbounded.

6. Fixed-length local stability (elementary)

Lemma 4 (no novelty claim). Fix n and $q_0 \in (0, 1)$. Suppose an optimizing distance polynomial P has a gap of at least $\gamma > 0$ below each other **distinct** distance polynomial at q_0. Then the same P uniquely optimizes among distance polynomials whenever

$$|q - q_0| < \gamma/(4n), \qquad q \in [0, 1]. \tag{5}$$

Proof. Couple deletion masks at q, q' via the same independent uniforms. For any fixed input word, the output laws differ in total variation by at most $n|q - q'|$. By the triangle inequality, each pair distance varies by at most $2n|q - q'|$. A difference of two pair distances thus varies by at most $4n|q - q'|$; strict positivity of the initial gap persists. □

This lemma concerns a *fixed length only*. It does not imply a recurrence in n.

7. Failed proof route and open-ended research question

The principal proposed generalization was

$$\limsup_{n \to \infty} S(n) = \infty \ ? \tag{6}$$

The investigation considered k-deck agreement, Thue–Morse-type pairs, high-deletion-order asymptotics, and insertion of common symbols to grow extremal pairs. The deck constructions can make particular pairs hard to distinguish, but do **not** by themselves show these pairs minimize over *all* competitors. Local fixed-length stability also fails to provide a word-length recurrence. Finally, the direct same-parameter insertion recurrence is refuted by Proposition 3. None of this proves or disproves (6). The outcome of (6) is **UNPROVED IN THIS INVESTIGATION**, without asserting its status in the wider literature.

Earlier exploratory reports included length-nine-through-twelve envelopes and length-sixteen high-deletion data. These are **not accepted results of this v0.2 package** and remain outside its claims and certificates.

8. Reproducibility, independent checks, and audit boundaries

Run the one-shot Windows script from the extracted ZIP root:

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File .\run_audit_windows.ps1
```

The script verifies SHA-256 for every shipped file *before* running any verifier; output files are written only under `audit_output/`, never over the distributed evidence. The two primary scripts verify the $n = 4, 5$ certificate and insertion counterexample, and the $n = 6, 7, 8$ complete lower envelopes. A third script, `code/independent_mask_audit.py`, independently counts deletion masks using dictionaries and recomputes 12 rational-parameter exhaustive TV minima. The Python code uses the standard library only; no GPU, network service, or external package is required.

The independent AI auditor of v0.1 found the mathematics reproducible but noted a gap in the inference of Proposition 2, overly strong titular wording, a pair-versus-interval count ambiguity, and output-path inconsistency. All were addressed in v0.2. The author subsequently supplied a **PASS** outcome for the independent AI audit of the finite-length results and a packaging-audit PASS for v0.2.1. The author also reported the Windows PowerShell success line `ALL PASS: HASH + FINITE PROOFS + INDEPENDENT TV CHECK` on November 2, 2026. These supplied outcomes are distinguished from the release-preparation reruns recorded in the accompanying verification logs. A complete external-auditor transcript and a complete Windows execution transcript are not included in the supplied source ZIP. Historical audit documents and verifier console labels are preserved as records of their original stage; their old HOLD labels do not describe the present bibliographic version.

Claim status: `FINITE-LENGTH RESULTS PASS / FAILED INSERTION STRATEGY / GENERAL CONJECTURE UNPROVEN / NOVELTY INCONCLUSIVE / NOT HUMAN PEER-REVIEWED`.

References

[1] OpenAI (2026), *Quantitative lower bounds for trace reconstruction*, OpenAI Math family 122. https://github.com/openai/math/blob/main/lean/docs/122.md.

[2] J. Chrisnata, H. M. Kiah, S. Rao, A. Vardy, E. Yaakobi, and H. Yao (2019), *On the Number of Distinct k-Decks: Enumeration and Bounds*, ISCIT 2019. https://doi.org/10.1109/ISCIT.2019.8905191.

[3] R. Golm, M. Nahvi, R. Gabrys, and O. Milenkovic (2022), *The Gapped k-Deck Problem*. https://arxiv.org/abs/2201.12671.

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